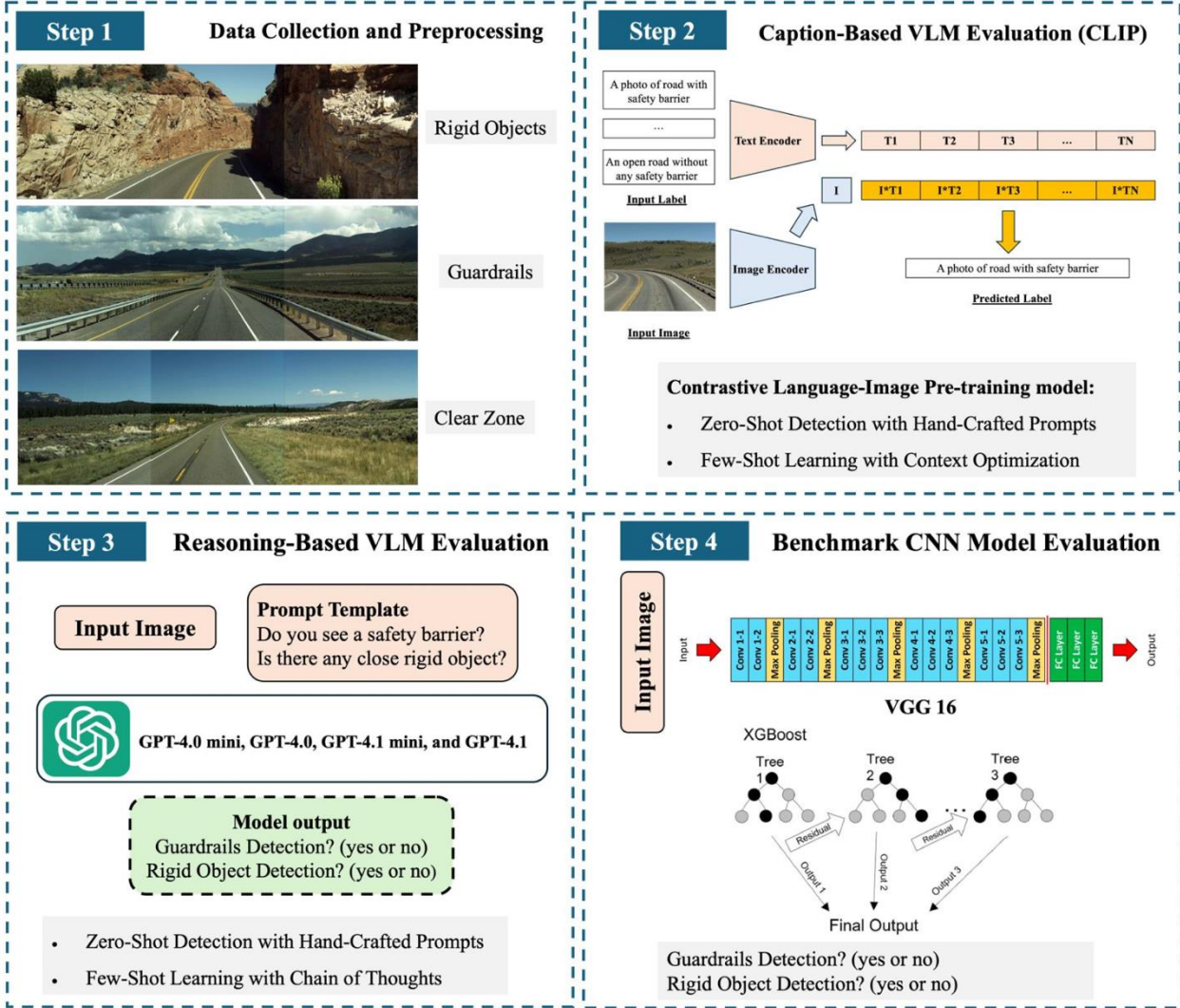




Step 3

Reasoning-Based VLM Evaluation

Input Image

Prompt Template

Do you see a safety barrier?

Is there any close rigid object?

GPT-4.0 mini, GPT-4.0, GPT-4.1 mini, and GPT-4.1

Model output

Guardrails Detection? (yes or no)

Rigid Object Detection? (yes or no)

- Zero-Shot Detection with Hand-Crafted Prompts
- Few-Shot Learning with Chain of Thoughts

Step 4

Benchmark CNN Model Evaluation

Input Image

Conv 1-1Conv 1-2Max PoolingConv 2-1Conv 2-2Max PoolingConv 3-1Conv 3-2Conv 3-3Max PoolingConv 4-1Conv 4-2Conv 4-3Max PoolingConv 5-1Conv 5-2Conv 5-3Max PoolingFC LayerFC Layer

VGG 16

XGBoost

Tree 1

Tree 2

Tree 3

Residual

Residual

Residual

Output 1

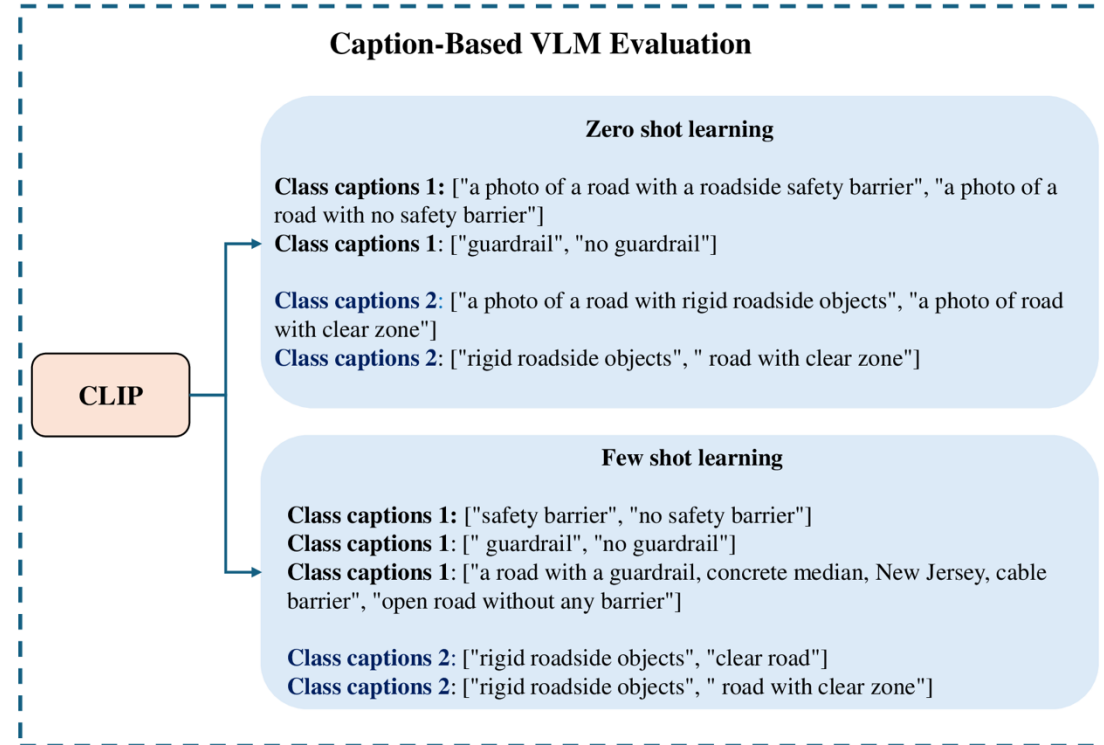
z indvO

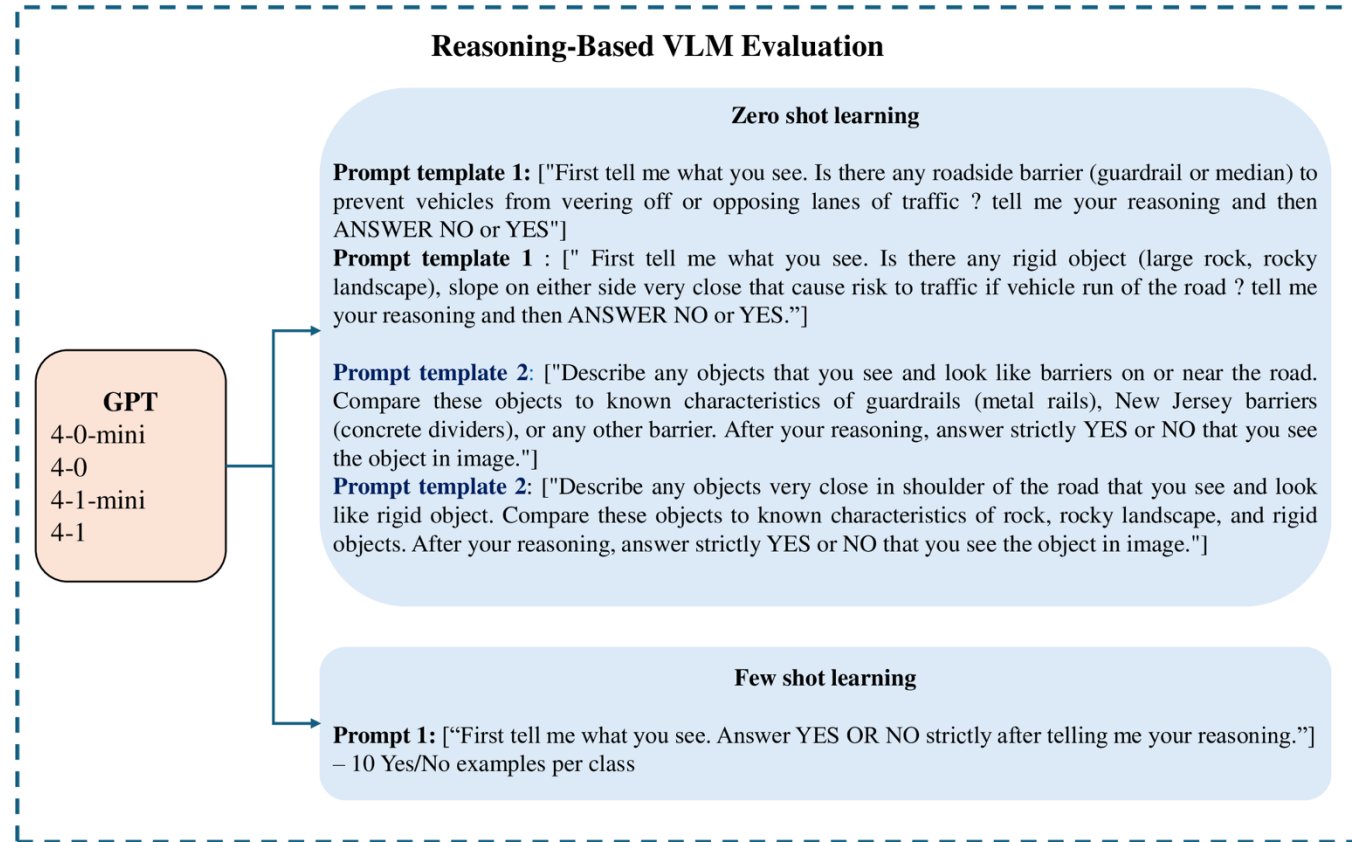
Output 3

Final Output

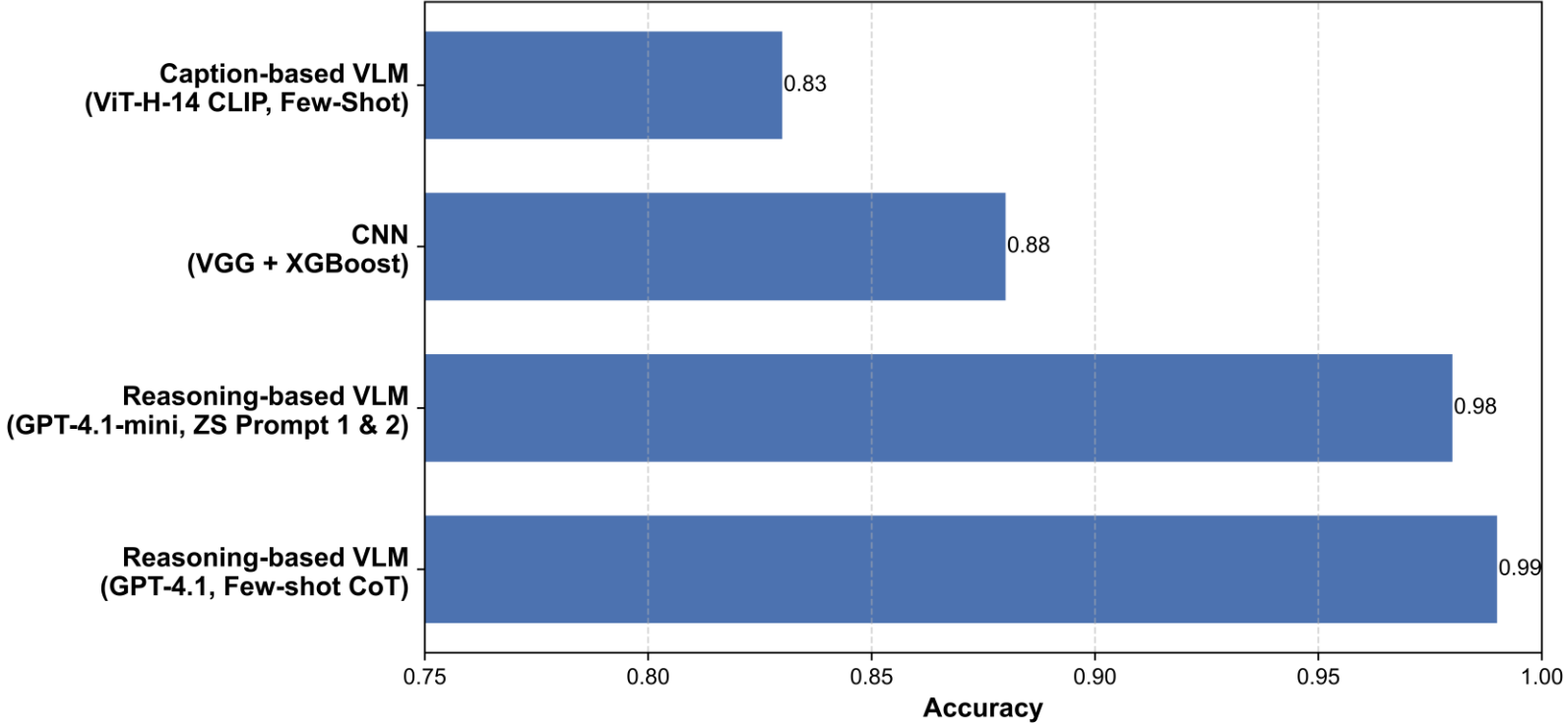
Guardrails Detection? (yes or no)

Rigid Object Detection? (yes or no)

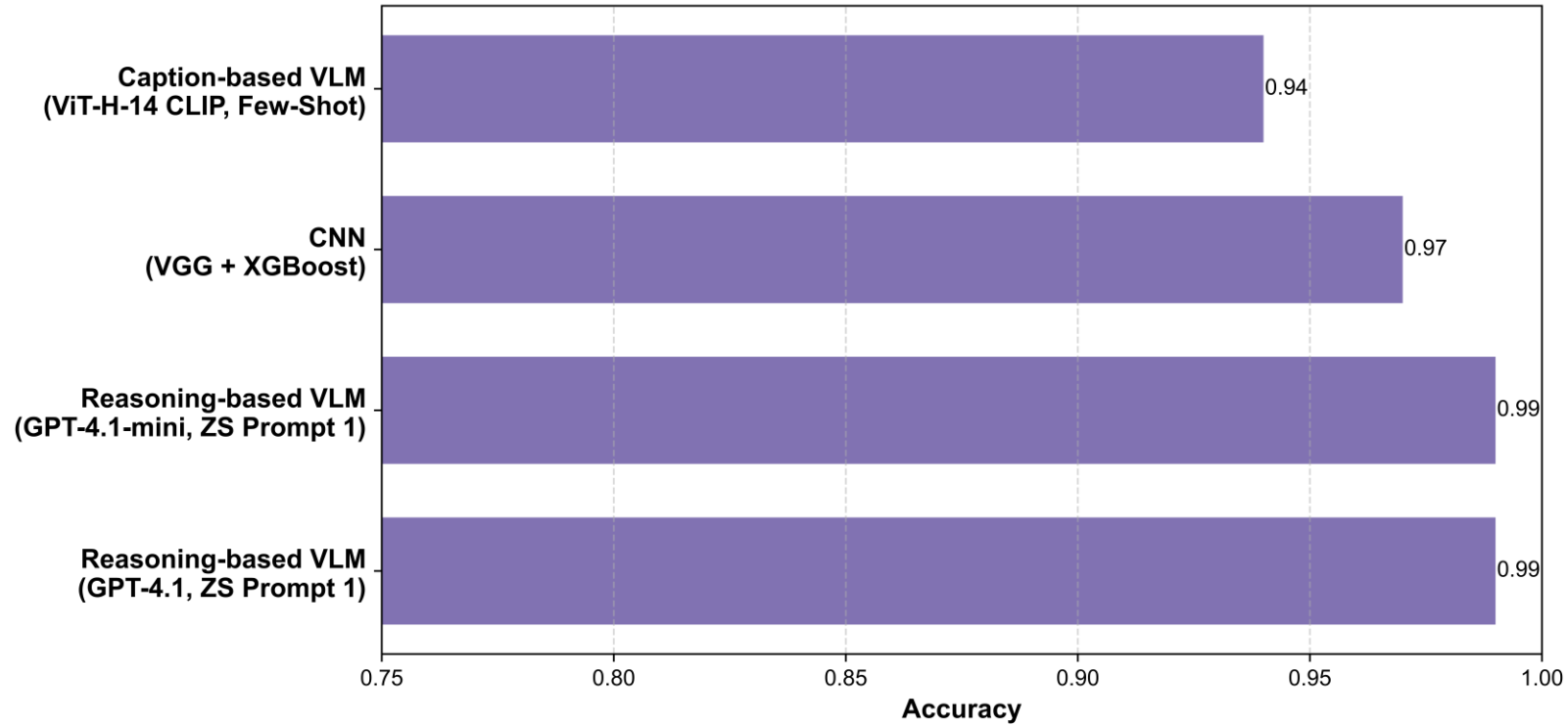


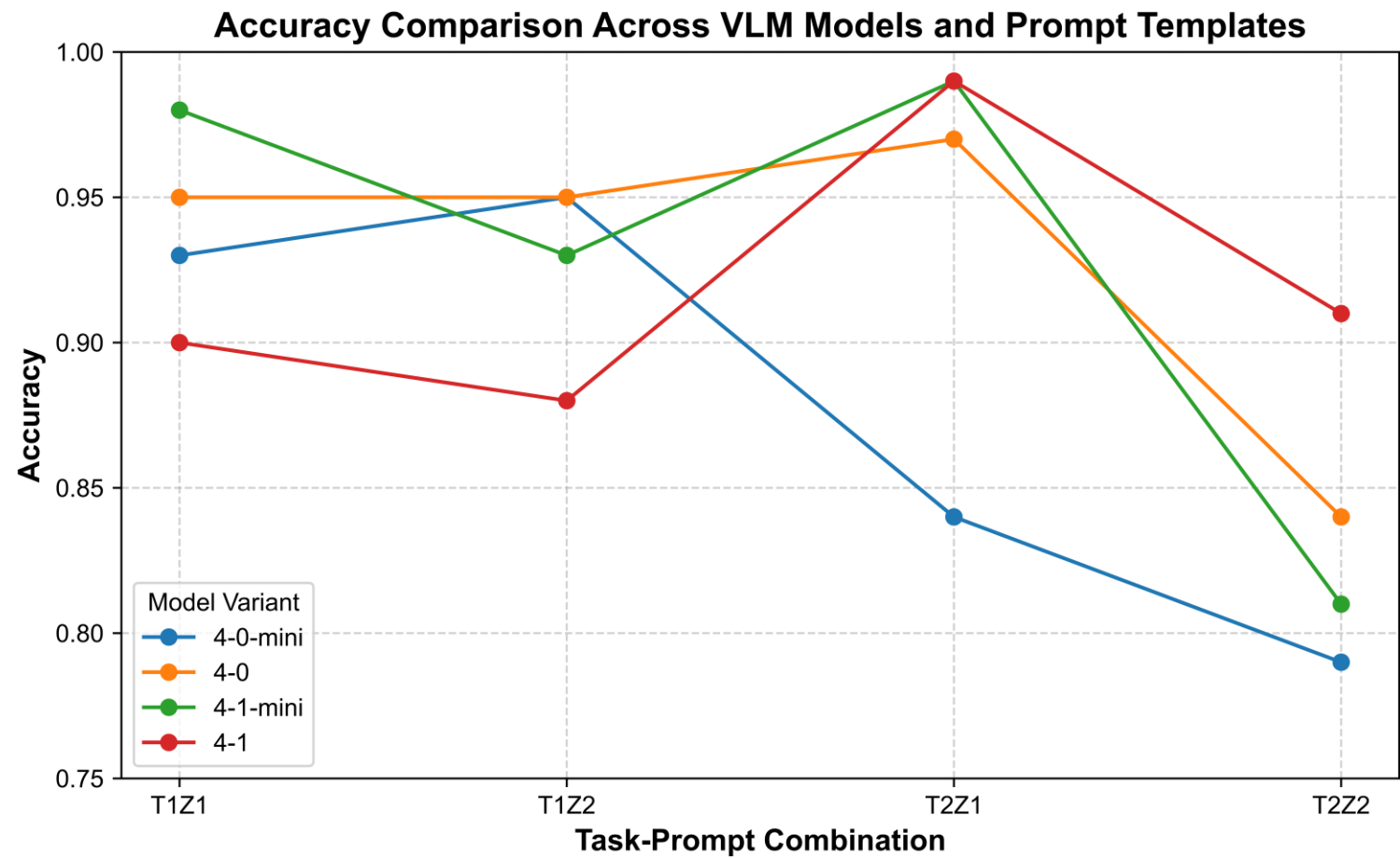


Best Model Performance on Safety Barrier Detection



Best Model Performance on Rigid Object Detection







4-1 mini Z1	In the image, I see a two-lane road winding through a rocky and bushy landscape. On the left side of the road, there is vegetation close to the road edge but no large rocks or steep slopes immediately next to the pavement. On the right side, there are large rocks and a rocky slope very close to the road edge. This side presents a rigid object hazard because a vehicle running off the road on that side could collide with the rocks or the slope, posing a risk to traffic. Therefore, considering the presence of rigid objects and a rocky slope very close on the right side of the road, the answer is: YES.
4-1 mini Z2	In the image, there are several objects very close to the shoulder of the road, particularly on the right side. These objects appear to be large, irregularly shaped, and have a rigid texture consistent with rocks or boulders. Their surfaces show characteristic signs of a rocky landscape, such as roughness, natural fractures, and earthy tones. They are solid, non-deformable, and appear integrated into the surrounding terrain, distinguishing them from organic or flexible objects like plants. The left shoulder of the road mostly shows dense vegetation without any similarly large rigid objects close to the road. The detailed rocky formations on the right side, scattered along the roadside, strongly resemble typical rock structures found in rocky landscapes. Given these observations, the rigid objects close to the shoulder of the road are consistent with rocks. Answer: YES
4-0 mini Z1	The image shows a winding road with rocky terrain on either side. The presence of large rocks and a rocky landscape suggests that there could be risks to traffic if a vehicle were to run off the road. Since there are noticeable rigid objects (rocks) nearby, the potential for danger is present. So, the answer is YES.
4-0 mini Z2	The objects close to the shoulder of the road appear to be rocky formations and boulders. When comparing these to known characteristics of rocks, they typically exhibit a hard, solid structure, often with irregular shapes and textures. Rocky landscapes often feature various colors and layers, which seem to be present in the image with the presence of terrain variations. Based on this reasoning, I would answer: YES.

	4-1 mini Z1	In the image, I see a two-lane road curving slightly to the right. On the right side of the road, there is a metal guardrail running alongside the edge of the pavement, which acts as a roadside barrier to prevent vehicles from veering off the road or heading into potentially dangerous areas such as the slope or drop-off. On the left side of the road, no barrier or guardrail is visible; it seems to be open and bordered by natural terrain with some trees. Since there is a metal guardrail on the right side of the road, this serves as a roadside barrier. Answer: Yes
	4-O mini Z1	I see a roadway that curves and has a gravel shoulder on one side. There appears to be no visible guardrail or median installed to prevent vehicles from veering off the road or crossing into oncoming traffic. The surface looks clear, and the absence of any barrier is notable. ANSWER: NO
	4-O mini Z2	I can't identify any barriers in the image, such as guardrails or New Jersey barriers, based on the description provided. NO.
	4-1 Z1	**What I see and my reasoning:** - The image shows a curved road with two lanes, divided by a visible double yellow line, indicating opposing traffic directions. - On the right side of the image (from the viewer's perspective), there is a metallic roadside barrier (guardrail) running along the curve of the road. - I do not see any median barrier physically separating the two traffic directions in the center, only the painted double yellow line.- On the left side, there is no visible barrier between the road and the roadside. ANSWER: YES